JACS
JOURNAL OF THE AMERICAN CHEMICAL SOCIETY
pubs.acs.org/JACS
Article

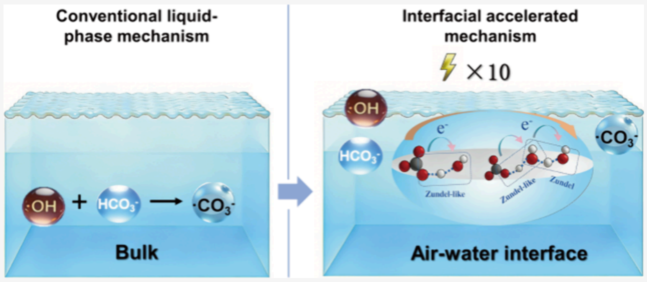
Redefining $\cdot CO_3^-$ Formation Chemistry: Zundel-like Switches Drive Carbonate-$\cdot$OH Interfacial Reactivity

Jiarong Liu, Xiaohua Yang, Jinkai Gu, Lili Qiu, Ling Liu, An Ning, Hao Li, Jinggang Lan,* Joseph S. Francisco,* and Xiuhui Zhang*

Cite This: *J. Am. Chem. Soc.* 2026, 148, 25501−25509
Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: The formation of carbonate radicals ($\cdot CO_3^-$) via carbonate-hydroxyl radicals ($\cdot$OH) reaction is the cornerstone of environmental oxidative cycles, yet its molecular mechanism has long been limited to homogeneous bulk-phase paradigms, a view that conflicts with enhanced reactivity in interfacial-rich systems. Characterizing these processes is hindered by the transience of $\cdot$OH, system heterogeneity, and the inability to resolve in situ pathways. Herein, we combine ab initio molecular dynamics and machine learning molecular dynamics to redefine $\cdot CO_3^-$ formation chemistry. We reveal that the gas−liquid interfacial reaction dominates $\cdot CO_3^-$ generation, mediated by two proton-coupled electron transfer pathways (concerted proton−electron transfer and stepwise proton-transfer followed by electron-transfer). Critical to this reactivity are Zundel/Zundel-like hydrogen-bonded configurations, which act as "molecular switches" to trigger rapid reactions, enabled by the intrinsic interfacial enrichment of $\cdot$OH (85.2%) and HCO_3^- (92.2%). The interfacial pathway outperforms bulk reactions in $\cdot CO_3^-$ formation, with (90 ± 6.13)% yield [vs (80 ± 8.94)% in bulk] and approximately 100-fold faster rate [$(1.15 \pm 0.01) \times 10^{11}$ M^{-1} s^{-1} vs $(9.63 \pm 0.03) \times 10^{8}$ M^{-1} s^{-1}], attributed to the partial solvation of $\cdot$OH at the interface. Additionally, $\cdot$OH reacts with bulk-phase CO_3^{2-} via heterogeneous electron transfer (bulk → interface), yielding a rate approximately 10-fold faster $\cdot CO_3^-$ formation than homogeneous bulk reactions. These findings challenge bulk-centric paradigms, establish the interface as the dominant $\cdot CO_3^-$ source, and provide actionable insights for optimizing advanced oxidation processes, water remediation, and catalyst design by leveraging interfacial microenvironments.



■ INTRODUCTION

The formation of carbonate radicals ($\cdot CO_3^-$) via carbonate-hydroxyl radicals ($\cdot$OH) reaction (involving HCO_3^- and CO_3^{2-}) is the linchpin of the environmental oxidative cycle.[1,2] It governs water purification via advanced oxidation processes (AOPs), self-purification of natural water, and formation of atmospheric secondary pollutants.[1,3−5] Over half a century, its understanding has relied entirely on the so-called "homogeneous bulk" paradigm, the assumption that all reactions occur uniformly in a well-mixed solution without interfacial or spatial heterogeneity.[6−9] Based on experimental data in homogeneous aqueous solutions, two core conclusions have been established: (i) carbonate acts as the efficient quencher of $\cdot$OH; (ii) as the concentration of carbonate increases, $\cdot$OH in the system is extensively consumed, leading to a predictable linear decrease in the system's oxidative capacity.[10] This paradigm is not only a classic theory in environmental chemistry textbooks but also directly informs practical work, including the design of bubbling AOPs reactors and the evaluation of pollution remediation in natural waters. However, accumulating evidence has revealed contradictions that the bulk paradigm cannot reconcile, indicating that real systems behave fundamentally differently from these classical predictions.

Recent studies across natural, engineered, and atmospheric environments consistently demonstrate this deviation. For natural water studies, Wang et al. used in situ laser-induced fluorescence to monitor lake surface water.[11] As the HCO_3^- concentration increased, the bulk paradigm predicted that $\cdot$OH quenching would significantly suppress phenol degradation efficiency. Yet, the observed phenol degradation rate increased instead, and the total concentration of oxidatively active species in the system did not exhibit the expected decline. In industrial bubbling AOPs, the addition of $NaHCO_3$ led to higher pollutant degradation efficiency.[4,12,13] This directly conflicts with the bulk-phase assumption that "high carbonate inhibits AOPs." Similarly, in atmospheric aerosol simulations,

Received: January 22, 2026
Revised: May 28, 2026
Accepted: June 1, 2026
Published: June 17, 2026

https://doi.org/10.1021/jacs.6c01510
J. Am. Chem. Soc. 2026, 148, 25501−25509

researchers have found that the concentration of $\cdot CO_3^-$ (the key product of the carbonate-·OH reaction) at the gas−liquid interface is 2 orders of magnitude higher than that predicted for the bulk-phase, directly implying that interfacial reactions dominate $\cdot CO_3^-$ formation.[3,14,15] Collectively, these findings challenge the universality of the bulk paradigm. Indeed, the surface layer of natural water bodies, aerosol interfaces, and gas−liquid contact zones in bubbling reactors share a unique feature: the existence of gas−liquid interface.[16] This interface exhibits abnormal physical-chemical properties that affect the reaction compared to bulk-phase reactions.[17,18] These observations further suggest that the key reaction between carbonate and ·OH may predominantly occur at the interface.

The microenvironmental differences between the gas−liquid interface and the bulk phase provide important clues to these contradictions.[19] On the one hand, unlike the disordered and random distribution of water molecules in the bulk phase, water molecules at the interface exhibit oriented ordering, forming an interfacial electric field that influences the reaction rate.[17,20,21] On the other hand, the HCO_3^- form of carbonate spontaneously migrates and accumulates at the interface driven by entropy, increasing the collision probability between reactants.[22,23] Meanwhile, ·OH shows significantly reduced solvation at the interface, resulting in more exposure of active sites and a longer residence time.[24,25] These well-established interfacial advantages indicate that the gas−liquid interface could entirely alter the conditions of carbonate-·OH reaction. Nevertheless, systematic investigations of this interfacial reaction remain scarce, primarily due to inherent limitations of conventional experimental techniques. Traditional methods (e.g., pulse radiolysis, laser flash photolysis) can quantify bulk-phase kinetics but are unable to resolve atomic-scale interfacial dynamics, capture transient microenvironmental effects, or map reactants' spatial distribution and solvation states at the gas−liquid boundary. As a result, the behavioral differences of ·OH and carbonate at the interface have not been quantitatively defined, nor has the specific impact of the interfacial environment on the reaction kinetics and product pathways been fully resolved. Consequently, the core question of whether the interfacial carbonate-·OH reaction is the root cause of the observed contradictions has remained speculative, leaving the interfacial mechanism insufficiently understood.

To address this knowledge gap, this study employs a cutting-edge combination of ab initio molecular dynamics (AIMD) and machine learning (ML) to build high-precision potential energy surfaces that overcome AIMD's time-scale limitations. We have investigated the reaction dynamics of ·OH with pH-dependent carbonate species (HCO_3^-/CO_3^{2-}) at the gas−liquid interface, aiming to redefine $\cdot CO_3^-$ formation chemistry. We also resolve the proton-coupled electron transfer (PCET) mechanisms and identify the key "molecular switches" driving rapid $\cdot CO_3^-$ generation. Most importantly, a systematic and quantitative comparison between homogeneous bulk solutions and gas−liquid interfacial systems is provided, yielding new understanding of oxidation chemistry beyond the bulk paradigm.

■ THEORETICAL METHODS

Classical Molecular Dynamics Simulation

Using the umbrella sampling method and periodic boundary conditions in the NVT ensemble, we computed the free-energy distribution of the reactants in various locations via the GROMACS program.[26] A vacuum layer of 4 nm was added along the z-axis, and the water box size was fixed at a size of 2 × 2 × 2 nm, containing 267 water molecules. Nonbonded interactions were fitted using Lennard-Jones and Coulomb potentials, whereas electrostatic interactions were fitted using the Ewald method, with a cutoff of 0.5 nm.[27,28] There were 40 umbrella sample windows spaced 0.5 nm apart. The completed phase was run for 2 ns as the last source of free-energy data after pre-equilibration in each window, with the step size set to 2 fs. The Nose-Hoover thermostat was used to maintain the temperature at 300 K.[29] Water molecules employed the SPC/E force field,[30] while all reactants used the General Amber Force Field (GAFF).[31] The weighted histogram analysis method (WHAM) was used to determine the free energy in various windows.[32]

Born−Oppenheimer Molecular Dynamics Simulation

By using the quickstep module of the CP2K code,[33] the Born−Oppenheimer molecular dynamics (BOMD) simulation of ·OH and HCO_3^- was carried out using density functional theory (DFT) in conjunction with the Gaussian plane wave (GWP) approach in a 1.7 × 1.7 × 4.7 nm periodic box that included 128 water molecules. The Perdew−Burke−Ernzerhof (PBE) functional[34] with the rVV10 nonlocal van der Waals correction[35] was used to calculate the interaction and exchange of electrons. This combination is not only computationally efficient but also reliable for describing both the electronic structures and the energetics.[36] The Goedecker-Teter-Hutter (GTH) pseudopotential[37] was used to describe the nuclear electrons, whereas the triple Zeta-polarized TZVP plane wave basis set was used to represent the valence electrons. 520 Ry was chosen as the plane wave basis set's energy cutoff.

Molecular Dynamics Simulation Based on ML Force Field/ML Potential

The 3000 reaction structures and energy data obtained from the aforementioned BOMD simulation, 90% of which were used as the training set and 10% as the validation set, were utilized as a new data set for fine-tuning MACE-MP-0.[38] The water cluster box was maintained in accordance with the aforementioned BOMD simulation settings while the Atomic Simulation Environment (ASE) program was utilized in conjunction with the refined ML force field model for molecular dynamics simulations.[38] The Langevin thermostat was used to maintain the temperature of the NVT ensemble at 300 K,[39] the simulation step was set to 1 fs, and the entire period was 20 ps. We conducted extra single-point energy calculations on the structural points chosen from the reaction trajectory on CP2K in order to gain precise information, such as the spin and charge of the reactants, as the MLP approach is unable to provide electronic structure information. We calculated the spin population every 10 fs during the reaction using hybrid DFT with PBEh(α) and 40%-HFX based on the ML trajectory. Spin density, defined as the difference between the charge densities of spin-α and spin-β electrons, reflects the local distribution of unpaired electrons within a molecular system.[40,41] It serves as a key observable for quantifying radical character, where positive and negative values correspond to net excesses of spin-α and spin-β electrons, respectively. In our calculations, ·OH has a net spin density of 1 corresponding to a doublet state (multiplicity = 2). This indicates one unpaired electron and confirms ·OH as an oxygen-centered monoradical. Auxiliary density matrix method (ADMM) was utilized to speed up the computation, and the geometrical response basis valence set (ccGRB-T) was employed to describe the electrons.[42]

It should be noted that the present classical molecular dynamics simulations do not explicitly account for quantum nuclear effects (NQEs), which are known to accelerate proton-transfer processes in aqueous environments.[43,44] However, they primarily affect the absolute reaction rates while having a negligible impact on the relative rates between different reaction pathways and the corresponding branching ratios. This is because the tunneling and zero-point energy corrections are comparable for similar hydrogen-bonded proton-transfer processes in the same system. Furthermore, the net effect of NQEs on the thermodynamic properties such as reaction free energies is typically small due to compensating enthalpic

https://doi.org/10.1021/jacs.6c01510
J. Am. Chem. Soc. 2026, 148, 25501−25509

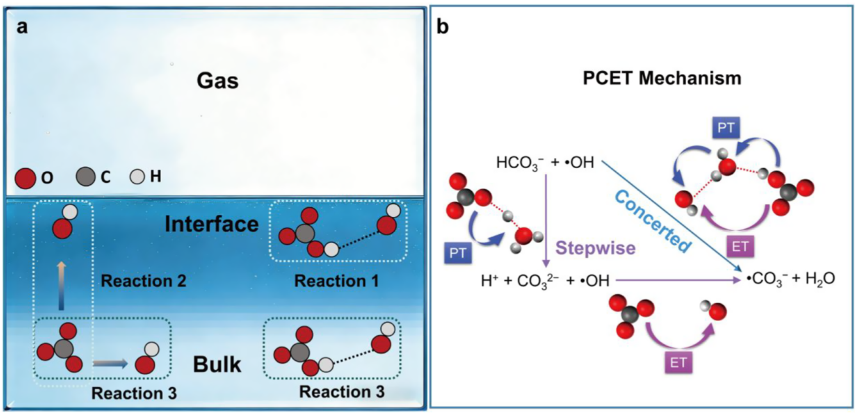


Figure 1. (a) Modeling of the interfacial and bulk-phase reaction systems. (b) Two PCET reaction mechanisms, namely, the CPET and PT-ET pathways. Color codes: O (red), C (grey), and H (white).

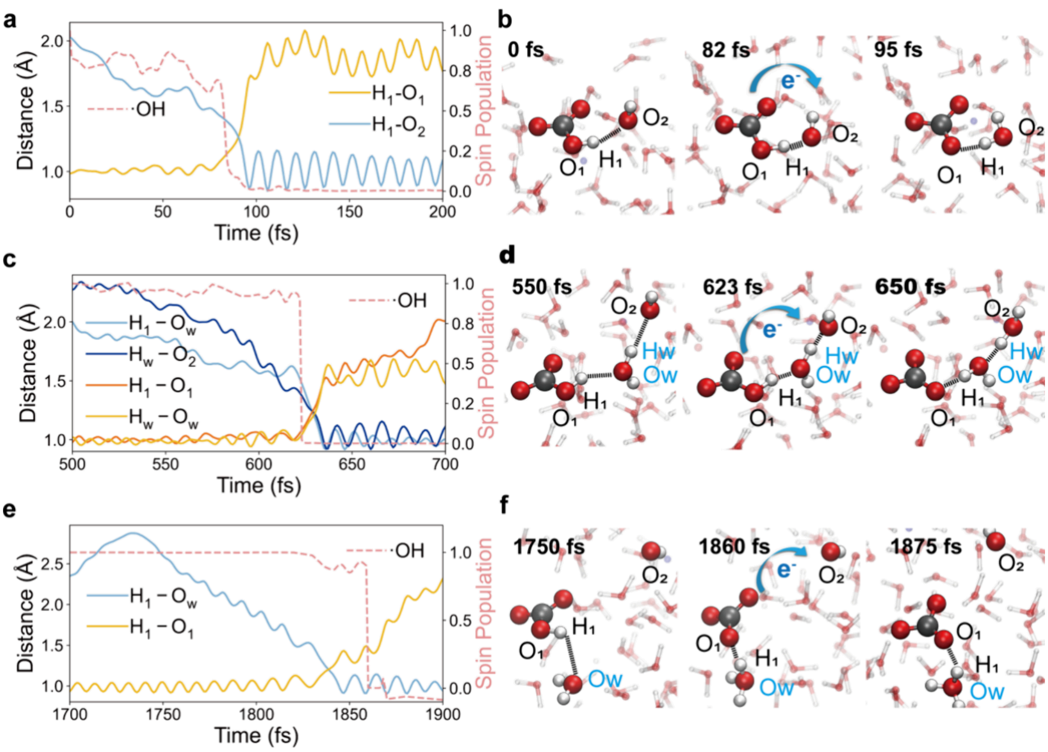


Figure 2. Characterization of different $\cdot CO_3^-$ mechanisms for the interfacial reaction between $\cdot OH$ and HCO_3^-. (a,c,e) Time-dependent changes in O−H distances and ·OH spin for different mechanisms: (a) CPET mechanism, (c) water-mediated CPET mechanism, and (e) PT-ET mechanism. (b,d,f) Structural snapshots of the corresponding mechanisms. Color codes: O (red), C (grey), and H (white).

and entropic contributions.[45,46] Thus, the qualitative conclusions regarding the reaction mechanism, the thermodynamic preference for interfacial localization of the reactants, and the role of key configurations remain valid. Future work employing path-integral molecular dynamics simulations will be conducted to obtain more precise absolute rate constants and further quantify the influence of the NQEs.

■ RESULTS

Explanation of Initial Configuration Settings for the Reaction

To address the long-standing bulk-phase paradigm that assumes carbonate-·OH reactions occur exclusively in homogeneous solutions, a critical prerequisite is to clarify the spatial distribution of reactants (·OH and carbonate) within the system. As emphasized in the introduction, this bulk-centric view fails to explain interfacial contradictions, and here we directly resolve this by mapping reactant locations. Figure S1 illustrates the movement of HCO_3^-/CO_3^{2-} and ·OH across the air−water interface and the corresponding Gibbs free-energy changes: HCO_3^- exhibits significant enrichment at the air−water interface (92.2%, corresponding to the lowest free energy), and ·OH also accumulates there (85.2%), while CO_3^{2-} preferentially resides in the bulk phase, consistent with previous experimental findings[23,47] but in stark contrast to the bulk paradigm's implicit assumption. It is worth emphasizing that the interfacial preference results obtained from the ML force field simulations further validate these findings, with excellent agreement observed (Figure S2). Based on this spatial characterization, to elucidate the interfacial reaction mecha-

https://doi.org/10.1021/jacs.6c01510
J. Am. Chem. Soc. 2026, 148, 25501−25509

nism and explicitly contrast interfacial and bulk behaviors, three distinct reaction systems were simulated (Figure 1a): (1) Interfacial carbonate-·OH reaction (·OH + HCO_3^-); (2) reaction between interfacial ·OH and bulk-phase CO_3^{2-}; (3) bulk-phase carbonate-·OH reactions (·OH + HCO_3^-/CO_3^{2-}). To comprehensively characterize the interfacial reaction mechanism, we designed initial reaction configurations considering two key factors: reactant distance (affecting the strength of intermolecular interactions) and relative orientation between the active site of ·OH (oxygen atom) and the reaction sites of HCO_3^- (e.g., hydroxyl oxygen, carbonate oxygen), altering the transition state stability. We constructed diverse spatial arrangements and simulated 22 trajectories for interfacial ·OH-HCO_3^- reactions (details in the Supporting Information).

Formation Mechanism of ·CO_3^- from ·OH and HCO_3^- at the Air–Water Interface

As illustrated in Figure 1b, two distinct PCET mechanisms[48] are revealed for the interfacial reaction between ·OH and HCO_3^-, namely, the concerted proton–electron transfer (CPET) and the stepwise proton-transfer followed by electron-transfer (PT-ET) pathways. Notably, the difference between these two mechanisms lies in that CPET involves the concerted transfer of an electron and a proton in a single elementary step, whereas PT-ET proceeds via a stepwise process with proton transfer preceding electron transfer.[49,50] Meanwhile, these mechanisms are observed not only at the interface but also in the bulk phase (Figure 1a), with the 22 simulated trajectories detailed in Figure S4a, as well as the corresponding reaction ratios are presented in Figure S4b and their differences to be elaborated in detail in the subsequent discussion. Proton and electron transfer events are presented in Figure S4b. Specifically, the CPET mechanism accounts for 45% of the reactions, while the PT-ET mechanism contributes 33%. Figure 2a–c depicts the changes in the O–H distances and the spin multiplicity of ·OH to illustrate the proton transfer (PT) and electron transfer (ET) processes, respectively. Taking the anhydrous CPET mechanism (Figure 2a) as an example, over time, the gradual increase in the O1–H1 distance within HCO_3^- (yellow solid line) and the gradual decrease in the H1–O2 distance (blue line) denote the PT process, while the progressive reduction in the spin multiplicity of ·OH (red dashed line) signifies the ET process. Ultimately, the ·CO_3^- is generated, and the corresponding structural evolution is illustrated in Figure 2b.

Both the interfacial reaction pathways of ·OH with HCO_3^- (CPET and PT-ET) belong to PCET,[48] but they differ in the timing of PT and ET. CPET involves near-simultaneous PT and ET via a distinct configurational arrangement, and PT-ET follows a clear stepwise sequence: PT occurs first and then ET. Notably, not only the interfacial bimolecular reactions (·OH-HCO_3^- in Figure 2a), but also the trimolecular reactions (e.g., ·OH-HCO_3^- with one water molecule in Figure 2b) and even tetramolecular reactions (·OH-HCO_3^- with two water molecules in Figure S9) require the prior formation of the aforementioned specific configurational arrangement to proceed. Such a configuration in the CPET mechanism is referred to as a hydrogen-bonded complex [HCO_3^-···$(H_2O)_n$···˙OH], and our simulation results show that once this complex forms, the reaction proceeds promptly, even from mismatched initial configurations. This confirms that this specific type of hydrogen-bonded configuration drives the CPET mechanism. Analysis of prereaction configurations reveals two scenarios for CPET: HCO_3^-···(·OH) forms a Zundel-like configuration (in Figure S8a), or HCO_3^-···H_2O···(·OH) forms dual Zundel-like configurations (in Figure S8b), which act as "molecular switches" to trigger rapid ·CO_3^- formation. These Zundel-like configurations possess the same O···H···O motif as the canonical Zundel structure.[51,52] However, they exhibit asymmetry because of the different chemical species involved. Therefore, they function as reactive intermediates rather than bulk solvation states. Prior studies have found that such configurations facilitate PT,[53,54] and our trajectories further confirm that they also promote ET.[55] For the PT-ET mechanism, all prereaction configurations adopt the motif HCO_3^-···$(H_2O)_n$ ($n \geq 0$). After this complex forms, a PT occurs from HCO_3^- to H_2O, generating CO_3^{2-} and H_3O^+. Approximately 50 fs later, an electron delocalizes from CO_3^{2-} to ·OH, yielding ·CO_3^- and OH^-.

To further clarify the nature of the aforementioned reaction processes, we calculated the projected density of states (PDOS) for reactants in aforementioned mechanisms. For the water-mediated CPET mechanism (Figure 3a–c), the initial energy gap between HCO_3^- and ·OH is 1.30 eV. However, the strong highest occupied molecular orbital-lowest unoccupied molecular orbital (HOMO–LUMO) overlap (Figure 3b) reduces it to 1.10 eV, explaining why the formation of Zundel-like structures and the involvement of water could promote the reaction. For the direct CPET mechanism (Figure 3d), even though the initial energy gap is as high as 3.7 eV, the reaction can still occur after the formation of Zundel-like structures—further highlighting the key role of Zundel-like structures. For PT-ET (Figure 3e), the initial gap is 1.8 eV; after PT converts HCO_3^- to CO_3^{2-}, the gap narrows to 0.3 eV, enabling easy ET.

Formation Mechanism of ·CO_3^- between Interfacial ·OH and Bulk-Phase CO_3^{2-}

Furthermore, we extend our study to account for pH, a critical environmental factor that regulates carbonate speciation. Generally, HCO_3^- undergoes pH-dependent dissociation ($HCO_3^- \rightleftharpoons CO_3^{2-} + H^+$, $pK_a \approx 10.3$ at 25 °C), leading to dynamic shifts in the HCO_3^-/CO_3^{2-} ratio: HCO_3^- dominates at pH 6.5–10.3, while CO_3^{2-} becomes prevalent at pH > 10.3. This suggests that real environments feature pH-driven reaction process. Leveraging the bulk-dominant distribution of CO_3^{2-}, the reaction between interfacial ·OH and bulk CO_3^{2-} was simulated. Figure 4a presents four reaction trajectories of ˙OH with CO_3^{2-} at different distances ($D \approx 6$, 8, 9, and 10 Å). Across all four trajectories, the spin density of ·OH drops to 0 within 300 fs, indicating a rapid reaction process. Taking the specific long-distance reaction configuration (·OH–CO_3^{2-} at 9 Å) as an example, the PDOS analysis reveals that the origin of rapid electron transfer likely lies in a low-energy gap (0.75 eV in Figure 4b). The corresponding reaction proceeds exclusively via a single-electron transfer mechanism. Critically, this differs from the homogeneous electron transfer observed in bulk-phase ·OH–CO_3^{2-} simulations; instead, it constitutes heterogeneous electron transfer from bulk CO_3^{2-} to interfacial ·OH, with a reaction rate nearly 10-fold faster than that in the bulk, which is attributed to the enrichment of ·OH at the interface.

Our findings highlight that the gas–liquid interface acts as a "reaction hub" for carbonate–˙OH chemistry, accommodating pH-dependent HCO_3^-/CO_3^{2-} ratios via complementary

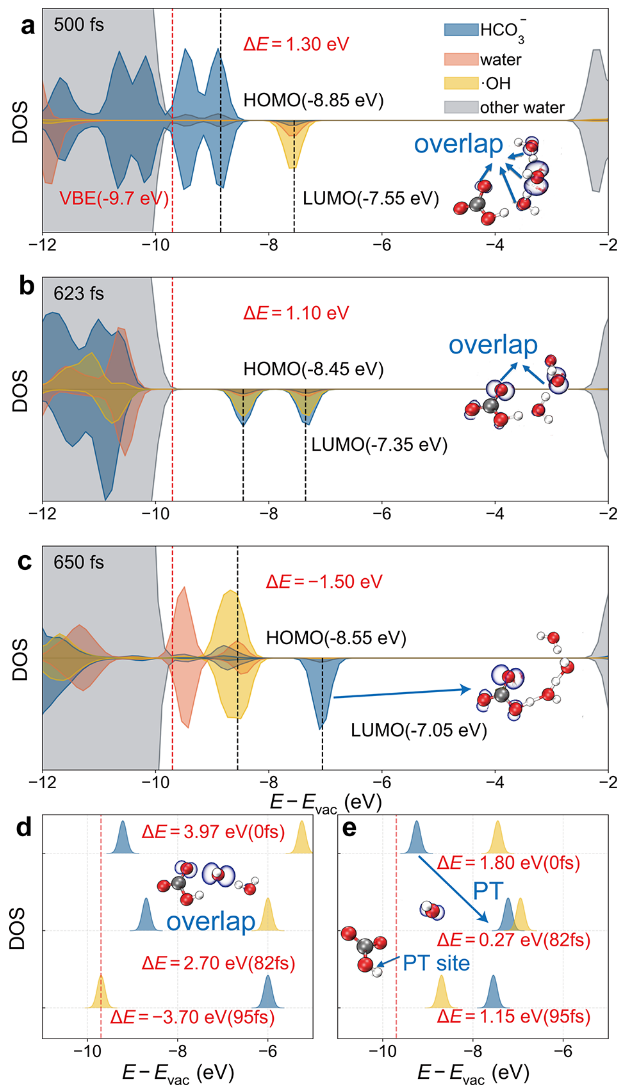


Figure 3. Energy-level changes of different mechanisms. (a,b,c) Water-mediated CPET mechanism at different time points. PDOS is shown, with labeled orbital species assignments and energy values. ΔE represents the energy gap. The inset shows the orbital localization of the unpaired electron, revealing the prereaction orbital overlap. (d) Direct CPET mechanism at different time points. (e) PT-ET mechanism at different time points. The figure marks the reduction of energy gap caused by the PT process. In all figures, the red dashed line represents the aqueous solution valence band edge (VBE).

mechanisms (CPET/PT-ET for $HCO_3^-$and heterogeneous single electron transfer for CO_3^{2-}) and sustaining high oxidative activity.

Comparison of Interfacial and Liquid-Phase $\cdot CO_3^-$ Formation Mechanisms

The significant concentration enrichment of HCO_3^- and ·OH at the gas−liquid interface, coupled with the rapid reaction kinetics from our simulations, points to interfacial reactions as a viable pathway. This stands in contrast to early pulse radiolysis studies in homogeneous liquid solutions, which underpinned the long-standing bulk-phase paradigm. To systematically compare similarities and differences between HCO_3^- and ·OH across distinct environments (interface vs bulk), we performed additional simulations in the liquid phase. The initial configurations were selected with reference to those of interfacial reactions, excluding two trajectories where the

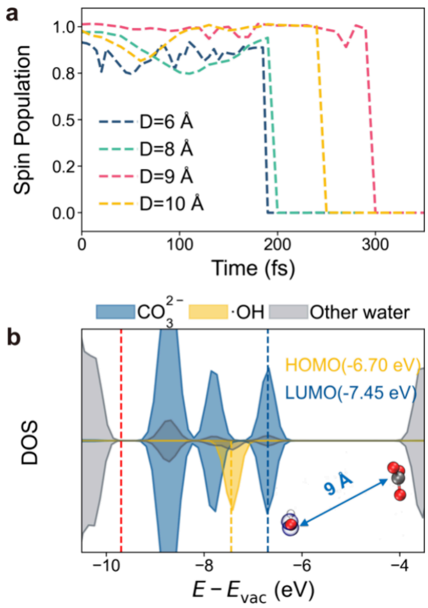


Figure 4. Characterization of the reaction mechanisms between interfacial ·OH and bulk-phase CO_3^{2-}. (a) ·OH spin for all trajectories. (b) Energy-level gap between CO_3^{2-} and ·OH in the prereaction structure at 9 Å, indicating that electrons can readily transfer from CO_3^{2-} to the ·OH radical. The red dashed line represents the aqueous solution valence band edge (VBE).

−OH group of HCO_3^- faces the gas phase (case E), resulting in 20 trajectories. Bulk-phase reactivity reached (80 ± 8.94)% [(60 ± 13.40)%: PT-ET, (20 ± 8.90)%: CPET], whereas the gas−liquid interface achieved a (90 ± 6.13)% $\cdot CO_3^-$ formation probability, 12.5 ± 10.80% points higher (Figure 5a). Notably, the proportions of the distinct PCET mechanisms differ at the interface as well. In particular, the CPET pathway accounts for (40 ± 10.40)% of the total mechanism at the interface in contrast to only (20 ± 8.90)% in the bulk phase. Kinetic analysis, deriving from ·OH spin dynamics and lifetime measurements (Figure S6), revealed interface rate constants up to 10^{11} s^{-1} (Table S1). Combined with the interfacial reactant enrichment, the reaction rate at the interface is more than 2 orders of magnitude faster than that in the bulk liquid [$\nu_{interface}$: $(1.15 \pm 0.01) \times 10^{11}$ M^{-1} s^{-1} vs ν_{bulk}: $(9.63 \pm 0.03) \times 10^{8}$ M^{-1} s^{-1}].

To uncover the origin of this rate difference, we compared the PDOS plots of reactions with matched initial configurations in both environments. As shown in Figure 5b, the prereaction energy gap between reactants is 2.0 eV in the bulk phase, significantly larger than the 1.3 eV gap at the interface. This energy alignment advantage explains the faster interfacial reaction process, directly validating the speculation that the interfacial microenvironment reshapes the reaction thermodynamics. It is worth emphasizing that the configurations in these two environments are similar and follow the same mechanism, as shown in Figure S10.

While numerous studies have established that interfacial environments accelerate reactions relative to the bulk, the underpinning mechanism remains debated. Our work resolves this ambiguity by identifying a dual driver: the formation of stable Zundel-like hydrogen-bonded structure in the prereaction configuration [$\cdot OH\cdots HCO_3^-\cdots(H_2O)_n$ ($n \geq 0$)] and the regulatory role of solvation extent—two factors tightly coupled to reaction feasibility. Figure S5 tracks the hydrogen-

https://doi.org/10.1021/jacs.6c01510
J. Am. Chem. Soc. 2026, 148, 25501−25509

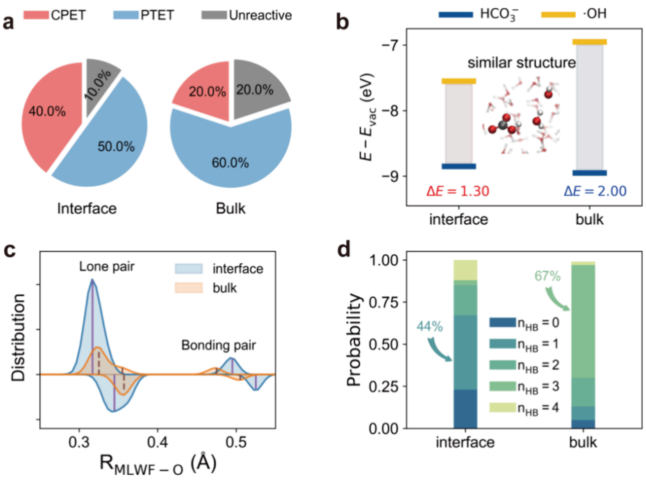


Figure 5. Comparison of $\cdot OH\text{-}HCO_3^-$ reaction characteristics: liquid phase vs interface. (a) Reaction probability statistics. (b) Energy-level information on water-involved CPET mechanisms in interfacial and liquid-phase environments. (c) Distributions of distances from the intramolecular oxygen of ·OH to the maximally localized Wannier centers in the interfacial and liquid-phase environments. In each panel, the vertical solid and dashed lines on the right indicate the average length of the covalent O−H bond, while the vertical solid and dashed lines on the left indicate the average distance from the intramolecular oxygen to the Wannier centers of lone and bonding pairs. The amphiphilic character of ·OH emerges from the comparison: an active site (oxygen or hydrogen) is hydrophobic (hydrophilic) if the separation between the lone pair and oxygen or between the bonding pair and hydrogen is shorter (longer) at the interface than in the liquid phase. (d) Hydrogen bond acceptor statistics for ·OH.

bond lifetimes in reactive configurations, linking the stable hydrogen-bond (a hallmark of Zundel-like structures) to the reaction success. Reactive systems require persistent hydrogen bonds: bulk-phase reactions demand longer lifetimes (0.876 ps) to form such structures, whereas interfacial reactions achieve stability with shorter lifetimes (0.424 ps). In contrast, nonreactive configurations in both environments lack stable hydrogen bonds (lifetimes < 0.100 ps). This confirms that interface−bulk differences modulate $\cdot CO_3^-$ formation probability by governing the ability to form Zundel-like hydrogen-bonded geometries, with the interface inherently favoring this reactive arrangement.

To dissect how solvation and Zundel-like structure formation shape reaction mechanisms, we compared the interfacial and bulk-phase trajectories with the matched initial configurations and quantified their reactivity. At the interface, ·OH exhibits enhanced mobility due to partial solvation. This flexibility allows ·OH to readily adjust its position relative to HCO_3^-, facilitating the formation of Zundel-like structures across different separations. In stark contrast, bulk-phase ·OH is fully solvated, drastically reducing its mobility.

We further elucidated the solvation effect both qualitatively and quantitatively by linking the electronic structure of ·OH to its hydrogen-bonding behavior and Zundel-like structure formation. Qualitatively, ·OH features one unpaired electron and one lone electron pair [as shown by maximally localized Wannier functions (MLWFs) in Figure 5c], making its proton positively charged (prone to donating to adjacent water molecules) and its oxygen electronegative (prone to accepting hydrogen bonds from adjacent water molecules due to the lone pair). Since hydrogen bonds are electrostatic interactions, their acceptor capacity can be approximated by the distance between the lone pair (negative charge center) and positive nuclei, directly reflecting local positivity. Analysis of these electron pair-nucleus distances (Figure 5c) reveals a key difference: compared to the bulk, interfacial ·OH has a stronger proton-donor ability but a weaker oxygen-acceptor ability. Quantitatively, this electronic signature makes interfacial ·OH hydrophobic, favoring monodentate hydrogen-bonded configurations (44% in Figure 5d) that support a rapid reaction. In the bulk phase, the fraction of ·OH acting as the hydrogen-bond acceptor prereaction further confirms this trend: ·OH predominantly forms tridentate hydrogen-bonded structures (67% in Figure 5d), stabilized by an abundant surrounding solvent. This fully solvated configuration is thermodynamically stable but kinetically inert—its rigidity hinders relaxation into the Zundel-like reactive geometry, forcing bulk reactions to rely more on PT from HCO_3^- than ET. Therefore, the different degrees of solvation of ·OH in the bulk liquid and at the interface not only explain the difference in reaction rates but also further account for the distinct proportions of the configuration-dependent CPET and PT-ET mechanisms.

Notably, the simulated reaction between ·OH and CO_3^{2-} is less sensitive to environmental conditions, an intriguing contrast to the $\cdot OH-HCO_3^-$ system. All reactive trajectories in both bulk and interfacial environments complete within 300 fs, with the electron transfer time weakly depending on reactant distance. However, the underlying mechanism differs sharply between the two environments. The reaction of bulk-phase ·OH with CO_3^{2-} follows a homogeneous electron transfer pathway. In contrast, the reaction between interfacial ·OH and bulk CO_3^{2-} proceeds via heterogeneous electron transfer, from bulk CO_3^{2-} to interfacial ·OH, yielding a reaction rate nearly 10-fold faster than that in the bulk.

■ CONCLUSIONS

In this work, we present a comprehensive theoretical investigation of the $\cdot CO_3^-$ formation via the carbonate-·OH reaction underpinning key environmental oxidative cycles, one that not only resolves the molecular-level reaction mechanism but also revolutionizes the decade-old "bulk-dominant" framework for this pivotal process. By combining ML-accelerated ab initio molecular dynamics, we overcome the intrinsic time-scale limitations of AIMD while surmounting experimental limitations that hinder atomic-scale tracking of interfacial dynamics, systematically exploring feasible reaction trajectories between ·OH and pH-dependent carbonate species (HCO_3^-/CO_3^{2-}) at the gas−liquid interface. Our simulations reveal that the interface serves as a central hub for $\cdot CO_3^-$ formation, driven by the intrinsic interfacial propensity of ·OH (85.2%) and HCO_3^- (92.2%). Two coupled electron transfer pathways (PT-ET and CPET) proceed through Zundel/Zundel-like hydrogen-bonded configurations between reactants, which act as "molecular switches" initiating the rapid reaction. The interfacial mechanism is fundamentally distinct from the bulk-phase reaction, which suffers from strong solvation-induced rigidity and limited access to reactive configurations. The interfacial ·OH exhibits partial, hydrophobic solvation, primarily monodentate hydrogen bonding, in contrast to rigid Eigen-like solvation in the bulk. This solvation structure enhances molecular mobility, promotes Zundel-like "switch," and increases $\cdot CO_3^-$ formation probability by 12.5 ±

https://doi.org/10.1021/jacs.6c01510
J. Am. Chem. Soc. 2026, 148, 25501−25509

10.80%, leading to a reaction rate nearly 2 orders of magnitude higher than that in homogeneous solution ($(1.15 \pm 0.01) \times 10^{11}$ M^{-1} s^{-1} vs $(9.63 \pm 0.03) \times 10^{8}$ M^{-1} s^{-1}). The formed $\cdot CO_3^-$ can readily oxidize a wide range of organic compounds, particularly S- and N-containing phenolics, thereby explaining why elevated carbonate concentrations do not suppress oxidative activity in real environments, contrary to bulk-phase predictions. We further identify a complementary heterogeneous electron-transfer pathway between interfacial $\cdot OH$ and bulk-dominant CO_3^{2-}, which proceeds nearly 10 times faster than bulk electron transfer. These findings establish a new interfacial mechanism for $\cdot CO_3^-$ formation that resolves the contradiction between bulk-phase predictions and environmental observations, elucidate the role of Zundel-like PCET and partial solvation in governing interfacial reactivity, and provide actionable insights for optimizing bubbling AOPs reactors, water remediation, and catalytic design. Overall, this work redefines the carbonate-$\cdot OH$ reaction as an intrinsically interfacial process and offers a general framework for understanding and exploiting interfacial redox chemistry in complex environmental systems.

■ ASSOCIATED CONTENT

Data Availability Statement

Data generated and analyzed for this study that are not included in this article and its Supporting Information are available at https://github.com/s1lencewill/Carbonate--OH-Interfacial-Reactivity.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.6c01510.

Detailed computational methods, additional figures (including the interfacial preference of reactants, initial configuration setup, and reaction mechanisms), and calculated table of reaction rates (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Jinggang Lan − *Faculty of Synthetic Biology, Shenzhen University of Advanced Technology, Shenzhen 518038, China*; orcid.org/0000-0001-6353-2539; Email: jinggang.lan@suat-sz.edu.cn

Joseph S. Francisco − *Department of Earth and Environmental Science and Department of Chemistry, University of Pennsylvania, Philadelphia, Pennsylvania 19104-6316, United States*; orcid.org/0000-0002-5461-1486; Email: frjoseph@sas.upenn.edu

Xiuhui Zhang − *State Key Laboratory of Environment Characteristics and Effects for Near-space, Beijing Key Laboratory of Intelligent Molecular Materials and High-throughput Manufacturing, Key Laboratory of Cluster Science, Ministry of Education of China, School of Chemistry and Chemical Engineering, Beijing Institute of Technology, Beijing 100081, China*; orcid.org/0000-0001-9570-7882; Email: zhangxiuhui@bit.edu.cn

Authors

Jiarong Liu − *State Key Laboratory of Environment Characteristics and Effects for Near-space, Beijing Key Laboratory of Intelligent Molecular Materials and High-throughput Manufacturing, Key Laboratory of Cluster Science, Ministry of Education of China, School of Chemistry and Chemical Engineering, Beijing Institute of Technology, Beijing 100081, China; School of Resources, Environment and Architectural Engineering, Chifeng University, Chifeng 024000, China*

Xiaohua Yang − *State Key Laboratory of Environment Characteristics and Effects for Near-space, Beijing Key Laboratory of Intelligent Molecular Materials and High-throughput Manufacturing, Key Laboratory of Cluster Science, Ministry of Education of China, School of Chemistry and Chemical Engineering, Beijing Institute of Technology, Beijing 100081, China*; orcid.org/0000-0002-3435-6473

Jinkai Gu − *State Key Laboratory of Environment Characteristics and Effects for Near-space, Beijing Key Laboratory of Intelligent Molecular Materials and High-throughput Manufacturing, Key Laboratory of Cluster Science, Ministry of Education of China, School of Chemistry and Chemical Engineering, Beijing Institute of Technology, Beijing 100081, China*

Lili Qiu − *State Key Laboratory of Environment Characteristics and Effects for Near-space, Beijing Key Laboratory of Intelligent Molecular Materials and High-throughput Manufacturing, Key Laboratory of Cluster Science, Ministry of Education of China, School of Chemistry and Chemical Engineering, Beijing Institute of Technology, Beijing 100081, China*; orcid.org/0000-0001-9693-4032

Ling Liu − *State Key Laboratory of Environment Characteristics and Effects for Near-space, Beijing Key Laboratory of Intelligent Molecular Materials and High-throughput Manufacturing, Key Laboratory of Cluster Science, Ministry of Education of China, School of Chemistry and Chemical Engineering, Beijing Institute of Technology, Beijing 100081, China*; orcid.org/0000-0003-3719-4083

An Ning − *State Key Laboratory of Environment Characteristics and Effects for Near-space, Beijing Key Laboratory of Intelligent Molecular Materials and High-throughput Manufacturing, Key Laboratory of Cluster Science, Ministry of Education of China, School of Chemistry and Chemical Engineering, Beijing Institute of Technology, Beijing 100081, China*

Hao Li − *Laboratory of Atmospheric Environment and Pollution Control, Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences, Beijing 100085, China; University of Chinese Academy of Sciences, Beijing 100049, China*; orcid.org/0000-0002-1002-6357

Complete contact information is available at: https://pubs.acs.org/10.1021/jacs.6c01510

Author Contributions

J.L., X.Y., and J.G. contributed equally. All authors approved the final version of the manuscript.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the National Science Fund for Distinguished Young Scholars (grant number 22225607) and the National Natural Science Foundation of China (grant numbers 22402128, 22476201, 22206017, 22376013, and 22306011).

■ REFERENCES

(1) Yang, J.; Zhao, Y.; Duan, M.; Deng, C.; Zhang, Y.; Lei, Y.; Li, J.; Song, W.; Chen, C.; Zhao, J. HCO_3^--mediated highly efficient photoelectrochemical dioxygenation of arylalkenes: triple roles of HCO_3^--derived radicals. *Energy Environ. Sci.* **2024**, *17* (1), 183−189.

(2) Jawad, A.; Chen, Z.; Yin, G. Bicarbonate activation of hydrogen peroxide: A new emerging technology for wastewater treatment. *Chin. J. Catal.* **2016**, *37* (6), 810−825.

(3) Liu, Y.; Li, X.; Ge, Q.; Fang, X.; Wang, T.; You, W.; Wang, W.; Xie, L.; Li, K.; Gong, K.; Yang, L.; Wang, R.; Wang, J.; Wang, L.; Ma, M.; Huang, T.; Fu, H.; Chen, J.; Dong, X.; Zhang, L. Carbonate radical ion as a key driver of rapid atmospheric sulfate formation. *npj Clim. Atmos. Sci.* **2025**, *8* (1), 45.

(4) Zhou, L.; Song, W.; Chen, Z.; Yin, G. Degradation of organic pollutants in wastewater by bicarbonate activated hydrogen peroxide with a supported cobalt catalyst. *Environ. Sci. Technol.* **2013**, *47*, 3833−3839.

(5) Huang, J.; Mabury, S. A. Steady-state concentrations of carbonate radicals in field waters. *Environ. Toxicol. Chem.* **2000**, *19* (9), 2181−2188.

(6) Buxton, G. V.; Greenstock, C. L.; Helman, W. P.; Ross, A. B. Critical review of rate constants for reactions of hydrated electrons, hydrogen atoms and hydroxyl radicals $\cdot OH/\cdot O^-$ in Aqueous Solution. *J. Phys. Chem. Ref. Data* **1988**, *17* (2), 513−886.

(7) Crittenden, J. C.; Hu, S.; Hand, D. W.; Green, S. A. A kinetic model for H_2O_2/UV process in a completely mixed batch reactor. *Water Res.* **1999**, *33* (10), 2315−2328.

(8) Liao, C.-h.; Gurol, M. D. Chemical oxidation by photolytic decomposition of hydrogen peroxide. *Environ. Sci. Technol.* **1995**, *29*, 3007−3014.

(9) Liao, C.-H.; Kang, S.-F.; Wu, F.-A. Hydroxyl radical scavenging role of chloride and bicarbonate ions in the H_2O_2/UV process. *Chemosphere* **2001**, *44*, 1193−1200.

(10) Shen, Q.; Song, X.; Fan, J.; Chen, C.; Guo, Z. Degradation of humic acid by UV/PMS: process comparison, influencing factors, and degradation mechanism. *RSC. Adv.* **2024**, *14* (32), 22988−23003.

(11) Wang, J.; Wang, K.; Zhang, L.; Guo, Y.; Guo, Z.; Sun, W.; Ye, Z.; Niu, J. Mechanism of bicarbonate enhancing the photodegradation of beta-blockers in natural waters. *Water. Res.* **2021**, *197*, No. 117078.

(12) Zhang, Y.; Peng, G.; Yan, Y.; Meng, X.; Gong, W. Highly efficient removal of organic pollutants with HCO_3^--enhanced ru(III)/NaClO process. *Int. J. Mol. Sci.* **2025**, *26* (2), 677.

(13) Fadaei, S.; Taheri, E.; Fatehizadeh, A.; Aminabhavi, T. M. New combination of pulsed light and iron (II) for carbonate radical production to enhanced degradation of bisphenol A: Parameter optimization and degradation pathway. *J. Environ. Manage.* **2022**, *322* (15), No. 116059.

(14) Wang, Y.; Zhou, C.; Chen, J.; Fu, Z.; Niu, J. Bicarbonate enhancing electrochemical degradation of antiviral drug lamivudine in aqueous solution. *J. Electroanal. Chem.* **2019**, *848*, No. 113314.

(15) Liu, Y.; Deng, Y.; Liu, J.; Fang, X.; Wang, T.; Li, K.; Gong, K.; Bacha, A. U.; Nabi, I.; Ge, Q.; Zhang, X.; George, C.; Zhang, L. A novel pathway of atmospheric sulfate formation through carbonate radicals. *Atmos. Chem. Phys.* **2022**, *22* (13), 9175−9197.

(16) Meng, Y.; Xia, Y.; Xu, J.; Zare, R. N. Spraying of water microdroplets forms luminescence and causes chemical reactions in surrounding gas. *Sci. Adv.* **2025**, *11*, No. eadt8979.

(17) Liu, Y.; Ge, Q.; Wang, T.; Zhang, R.; Li, K.; Gong, K.; Xie, L.; Wang, W.; Wang, L.; You, W.; Ruan, X.; Shi, Z.; Han, J.; Wang, R.; Fu, H.; Chen, J.; Chan, C. K.; Zhang, L. Strong electric field force at the air/water interface drives fast sulfate production in the atmosphere. *Chem.* **2024**, *10* (1), 330−351.

(18) Wilson, R. K.; Prophet, A. M. Chemical kinetics in microdroplets. *Annu. Rev. Phys. Chem.* **2024**, *75* (1), 185−208.

(19) Kong, X.; Ammann, M.; George, C.; Knopf, D. Atomic and molecular ions in atmospheric chemistry: interfacial reactivity, emerging mechanisms, and future perspectives. *J. Geophys. Res. Atmos.* **2025**, *130* (20), No. e2025JD044379.

(20) Zhang, J.; Liu, Z.; Zhu, C.; Wang, S.; Zhu, S.; Zhu, C.; Zhang, X. Ion-specific interfacial electric fields on water microdroplets for tuning menshutkin reactions. *J. Am. Chem. Soc.* **2025**, *147* (42), 38428−38434.

(21) Lee, K.; Mehrgardi, M. A.; Zare, R. N. Interfacial curvature, not simply size, controls spontaneous hydrogen peroxide formation in water microdroplets. *J. Am. Chem. Soc.* **2025**, *147* (36), 33240−33247.

(22) Martins-Costa, M. T. C.; Anglada, J. M.; Francisco, J. S.; Ruiz-Lopez, M. F. Reactivity of volatile organic compounds at the surface of a water droplet. *J. Am. Chem. Soc.* **2012**, *134* (28), 11821−11827.

(23) Du, H.; Liu, J.; Ozdemir, O.; Nguyen, A. V.; Miller, J. D. Molecular features of the air/carbonate solution interface. *J. Colloid Interface Sci.* **2008**, *318* (2), 271−277.

(24) Qiu, L.; Wei, Z.; Nie, H.; Cooks, R. G. Reaction acceleration promoted by partial solvation at the gas/solution interface. *ChemPlusChem.* **2021**, *86* (10), 1362−1365.

(25) Wei, Z.; Li, Y.; Cooks, R. G.; Yan, X. Accelerated reaction kinetics in microdroplets: overview and recent developments. *Annu. Rev. Phys. Chem.* **2020**, *71*, 31−51.

(26) Abraham, M. J.; Murtola, T.; Schulz, R.; Pall, S.; Smith, J. C.; Hess, B.; Lindahl, E. Gromacs: high performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX* **2015**, *1−2*, 19−25.

(27) Kawata, M.; Nagashima, U. Particle mesh Ewald method for three-dimensional systems with two-dimensional periodicity. *Chem. Phys. Lett.* **2001**, *340*, 165−172.

(28) Zhou, R.; Harder, E.; Xu, H.; Berne, B. J. Efficient multiple time step method for use with Ewald and particle mesh Ewald for large biomolecular systems. *J. Chem. Phys.* **2001**, *115* (5), 2348−2358.

(29) Evans, D. J.; Holian, B. L. The Nose−Hoover thermostat. *J. Chem. Phys.* **1985**, *83* (8), 4069−4074.

(30) Mark, P.; Nilsson, L. Structure and dynamics of the TIP3P, SPC, and SPC/E water models at 298 K. *J. Phys. Chem. A* **2001**, *105*, 9954−9960.

(31) Sprenger, K. G.; Jaeger, V. W.; Pfaendtner, J. The general AMBER force field (GAFF) can accurately predict thermodynamic and transport properties of many ionic liquids. *J. Phys. Chem. B* **2015**, *119* (18), 5882−95.

(32) Kumar, S.; Bouzida, D.; Swendsen, R. H.; Kollman, P. A.; Rosenberg, J. M. The Weighted Histogram Analysis Method for Free-Energy Calculations on Biomolecules. *I. The Method. J. Comput. Chem.* **1992**, *13* (8), 1011−1021.

(33) Kühne, T. D.; Iannuzzi, M.; Del Ben, M.; Rybkin, V. V.; Seewald, P.; Stein, F.; Laino, T.; Khaliullin, R. Z.; Schütt, O.; Schiffmann, F.; Golze, D.; Wilhelm, J.; Chulkov, S.; Bani-Hashemian, M. H.; Weber, V.; Borštnik, U.; Taillefumier, M.; Jakobovits, A. S.; Lazzaro, A.; Pabst, H.; Müller, T.; Schade, R.; Guidon, M.; Andermatt, S.; Holmberg, N.; Schenter, G. K.; Hehn, A.; Bussy, A.; Belleflamme, F.; Tabacchi, G.; Glöß, A.; Lass, M.; Bethune, I.; Mundy, C. J.; Plessl, C.; Watkins, M.; VandeVondele, J.; Krack, M.; Hutter, J. CP2K: an electronic structure and molecular dynamics software package-Quickstep: Efficient and accurate electronic structure calculations. *J. Chem. Phys.* **2020**, *152* (19), 194103.

(34) Ernzerhof, M.; Scuseria, G. E. Assessment of the Perdew−Burke−Ernzerhof exchange-correlation functional. *J. Chem. Phys.* **1999**, *110* (11), 5029−5036.

(35) Vydrov, O. A.; Van Voorhis, T. Nonlocal van der Waals density functional made simple. *Phys. Rev. Lett.* **2009**, *103* (6), No. 063004.

(36) Perdew, J. P.; Burke, K.; Ernzerhof, M. Generalized gradient approximation made simple. *Phys. Rev. Lett.* **1996**, *77* (8), No. 3865.

(37) Goedecker, S.; Teter, M.; Hutter, J. Separable dual-space Gaussian pseudopotentials. *Phys. Rev. B* **1996**, *54*, 1703−1710.

(38) Batatia, I.; Benner, P.; Chiang, Y.; Elena, A. M.; Kovács, D. P.; Riebesell, J.; Advincula, X. R.; Asta, M.; Avaylon, M.; Baldwin, W. J.; Berger, F.; Bernstein, N.; Bhowmik, A.; Bigi, F.; Blau, S. M.; Cărare, V.; Ceriotti, M.; Chong, S.; Darby, J. P.; De, S.; Della Pia, F.; Deringer, V. L.; Elijošius, R.; El-Machachi, Z.; Fako, E.; Falcioni, F.; Ferrari, A. C.; Gardner, J. L. A.; Gawkowski, M. J.; Genreith-Schriever, A.; George, J.; Goodall, R. E. A.; Grandel, J.; Grey, C. P.; Grigorev, P.;

Han, S.; Handley, W.; Heenen, H. H.; Hermansson, K.; Ho, C. H.; Hofmann, S.; Holm, C.; Jaafar, J.; Jakob, K. S.; Jung, H.; Kapil, V.; Kaplan, A. D.; Karimitari, N.; Kermode, J. R.; Kourtis, P.; Kroupa, N.; Kullgren, J.; Kuner, M. C.; Kuryla, D.; Liepuoniute, G.; Lin, C.; Margraf, J. T.; Magdău, I. B.; Michaelides, A.; Moore, J. H.; Naik, A. A.; Niblett, S. P.; Norwood, S. W.; O'Neill, N.; Ortner, C.; Persson, K. A.; Reuter, K.; Rosen, A. S.; Rosset, L. A. M.; Schaaf, L. L.; Schran, C.; Shi, B. X.; Sivonxay, E.; Stenczel, T. K.; Sutton, C.; Svahn, V.; Swinburne, T. D.; Tilly, J.; van der Oord, C.; Vargas, S.; Varga-Umbrich, E.; Vegge, T.; Vondrák, M.; Wang, Y.; Witt, W. C.; Wolf, T.; Zills, F.; Csányi, G. A foundation model for atomistic materials chemistry. *J. Chem. Phys.* **2025**, *163* (18), 184110.

(39) Davidchack, R. L.; Handel, R.; Tretyakov, M. V. Langevin thermostat for rigid body dynamics. *J. Chem. Phys.* **2009**, *130* (23), 234101.

(40) Verdaguer, M.; Robert, V. *Comprehensive Inorganic Chemistry II (Second ed.)*; Elsevier: Amsterdam, 2013.

(41) Ruiz, E.; Cirera, J.; Alvarez, S. Spin density distribution in transition metal complexes. *Coord. Chem. Rev.* **2005**, *249* (23), 2649−2660.

(42) Hehn, A.-S.; Sertcan, B.; Belleflamme, F.; Chulkov, S. K.; Watkins, M. B.; Hutter, J. Excited-state properties for extended systems: efficient hybrid density functional methods. *J. Chem. Theory* **2022**, *18* (7), 4186−4202.

(43) Hellström, M.; Ceriotti, M.; Behler, J. Nuclear quantum effects in sodium hydroxide solutions from neural network molecular dynamics simulations. *J. Phys. Chem. B* **2018**, *122* (44), 10158−10171.

(44) Ugur, B. E.; Webb, M. A. Nuclear quantum effects in molecular liquids across chemical space. *Nat. Commun.* **2025**, *16* (1), 5786.

(45) Markland, T. E.; Ceriotti, M. Nuclear quantum effects enter the mainstream. *Nat. Rev. Chem.* **2018**, *2* (3), No. 0109.

(46) Berrens, M. L.; Kundu, A.; Calegari Andrade, M. F.; Pham, T. A.; Galli, G.; Donadio, D. Nuclear quantum effects on the electronic structure of water and ice. *J. Phys. Chem. Lett.* **2024**, *15* (26), 6818−6825.

(47) Enami, S.; Sakamoto, Y. OH-radical oxidation of surface-active cis-pinonic acid at the air−water interface. *J. Phys. Chem. A* **2016**, *120* (20), 3578−3587.

(48) Hammes-Schiffer, S. Introduction: proton-coupled electron transfer. *Chem. Rev.* **2010**, *110* (12), 6937−6938.

(49) Warren, J. J.; Tronic, T. A.; Mayer, J. M. Thermochemistry of proton coupled electron transfer reagents and its implications. *Chem. Rev.* **2010**, *110*, 6961−7001.

(50) Weinberg, D. R.; Gagliardi, C. J.; Hull, J. F.; Murphy, C. F.; Kent, C. A.; Westlake, B. C.; Paul, A.; Ess, D. H.; McCafferty, D. G.; Meyer, T. J. Proton-coupled electron transfer. *Chem. Rev.* **2012**, *112* (7), 4016−4093.

(51) Biswas, R.; Voth, G. A. Role of solvation structure in the shuttling of the hydrated excess proton. *J. Chem. Sci.* **2017**, *129* (7), 1045−1051.

(52) Marx, D.; Tuckerman, M. E.; Hutter, J.; Parrinello, M. The nature of thehydrated excess proton inwater. *Nature* **1999**, *397*, 601−604.

(53) Suzuki, K.; Tachikawa, M.; Shiga, M. Temperature dependence on the structure of Zundel cation and its isotopomers. *J. Chem. Phys.* **2013**, *138* (18), 184307.

(54) Shin, I.; Park, M.; Min, S. K.; Lee, E. C.; Suh, S. B.; Kim, K. S. Structure and spectral features of $H^+(H_2O)_7$: Eigen versus Zundel forms. *J. Chem. Phys.* **2006**, *125* (23), 234305.

(55) Tian, Y.; Hong, J.; Cao, D.; You, S.; Song, Y.; Cheng, B.; Wang, Z.; Guan, D.; Liu, X.; Zhao, Z.; Li, X.-Z.; Xu, L.-M.; Guo, J.; Chen, J.; Wang, E.-G.; Jiang, Y. Visualizing Eigen/Zundel cations and their interconversion in monolayer water on metal surfaces. *Science* **2022**, *377* (6603), 315−319.

https://doi.org/10.1021/jacs.6c01510
J. Am. Chem. Soc. 2026, 148, 25501−25509